# AI-Powered Real-Time Smog Detection System Using Machine Learning and IoT Data

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***Abstract*—Smog is a severe environmental issue that affects air quality and public health. In this study, we propose a machine learning-based approach for real-time smog detection by integrating meteorological data with air quality indicators such as PM2.5, PM10, $NO_2$, and $O_3$. Using a dataset composed of real-time weather data from OpenWeatherMap API and air quality indices, we trained and evaluated three predictive models: Random Forest Regressor, XGBoost, and Long Short-Term Memory (LSTM). The Random Forest Regressor achieved the best performance with an $R^2$ score of 0.96 and a Mean Absolute Error (MAE) of 0.03. The results demonstrate that machine learning models can effectively predict smog levels and aid in environmental monitoring and public health protection.**

***Index Terms*—Smog Detection, Real-Time Monitoring, Random Forest, XGBoost, LSTM, AQI, IoT, Air Pollution.**

## I. Introduction

In recent years, the rapid industrialization, urbanization, and increasing vehicular traffic have severely impacted air quality in major cities, particularly across South Asia. One of the most visible and dangerous forms of air pollution is smog—a thick layer of harmful pollutants such as PM2.5, $NO_2$, and $O_3$, which severely threatens public health and environmental balance. While traditional air quality monitoring systems exist, they are often expensive, have limited coverage, and fail to provide real-time alerts, especially in developing regions like Pakistan. To address this, we designed and implemented an AI-powered, real-time smog detection system that leverages weather data and Air Quality Index (AQI) indicators. Our work includes the integration of real-time data from OpenWeatherMap and AQI sources, the application of machine learning models including Random Forest, XGBoost, and LSTM, and the deployment of a user-friendly dashboard for live smog predictions. The system is evaluated using metrics like MAE and $R^2$, with Random Forest achieving the highest performance. This research demonstrates a scalable, cost-effective, and intelligent approach to urban environmental monitoring.

### A. Problem Background

Air pollution continues to be the most pressing environmental risk to human health globally. According to the **2024 IQAir World Air Quality Report**, **99%** of the world's population lives in areas that fail to meet the World Health Organization (WHO) recommended air quality guideline for PM2.5 levels [1]. In 2024 alone, air pollution was linked to **8.1 million deaths**, with **58%** of these attributed specifically to ambient PM2.5 exposure [2], [3]. PM2.5, or fine particulate matter, has been shown to worsen cardiovascular and respiratory health, and is associated with developmental disorders, cognitive impairment, and mental health issues, particularly in children [4], [5]. The **2024 IQAir report** analyzed data from **over 40,000 monitoring stations** and **8,954 locations across 138 countries**, making it the most comprehensive global air quality survey to date [1]. **Pakistan** was ranked the **third most polluted country**, with an average PM2.5 concentration of **73.7 μg/m$^3$**, which is over **14 times higher** than WHO's safe limit [1]. Alarmingly, **only 17% of global cities** met WHO's annual PM2.5 guideline of 5 μg/m$^3$—though this was an improvement from 9% in 2023. These numbers reveal that the majority of the world's population is exposed to unsafe air, with developing nations and urban centers facing the most acute health risks. A critical concern raised in the report is that **only 21%** of the global population has access to **real-time, hyper-local air quality data** [1]. This creates a significant information gap in pollution awareness and response—especially in regions with high pollution but limited infrastructure. Addressing this gap through affordable, AI-powered smog detection systems is now a technological and ethical imperative for public health protection.

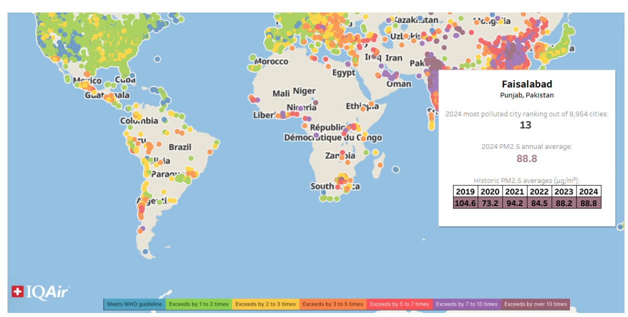



Fig. 1: Air Quality Report Background 2024

## II. Research Objectives

The primary objective of this research is to design and implement a real-time, AI-driven smog detection system that is both accurate and scalable. The specific goals of the study are as follows:

- To acquire and integrate real-time environmental data—such as meteorological parameters and air pollutant concentrations—using public APIs (e.g., OpenWeatherMap and AQI platforms).
- To preprocess the collected dataset through techniques like normalization, feature scaling, and feature selection, ensuring high-quality inputs for model training.
- To develop and evaluate the performance of multiple machine learning models, including Random Forest, XGBoost, and Long Short-Term Memory (LSTM), for smog level prediction.
- To assess model accuracy and reliability using standard evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination ($R^2$).
- To deploy the most effective model through an interactive, real-time dashboard for continuous air quality monitoring and public health awareness in urban environments.

## III. Research Questions

This study is guided by the following key research questions:

- How effectively can machine learning algorithms predict smog levels using real-time weather and air quality data?
- Among the selected models—Random Forest, XGBoost, and LSTM—which demonstrates the highest predictive performance in terms of accuracy, robustness, and generalizability?
- To what extent can a real-time, AI-powered dashboard enhance public awareness and support timely decision-making in response to poor air quality conditions?
- Is it feasible to develop a scalable and cost-efficient smog detection system using open-source data and lightweight technologies suitable for deployment in smart cities?

## IV. Methodology

This section outlines the systematic approach implemented to design, develop, and assess the proposed artificial intelligence-based smog detection framework. The research methodology comprises five distinct stages, each focusing on essential components of the workflow - ranging from initial problem analysis to final system implementation.

### A. Problem Identification

The initial stage consisted of an extensive examination of existing scholarly works to analyze contemporary technological solutions addressing smog-related challenges. Numerous academic publications were scrutinized, with particular emphasis on smog forecasting techniques, air quality assessment methods, and implementations of artificial intelligence and Internet of Things in environmental monitoring systems.

### B. Data Collection

During this phase, environmental data was acquired in real-time from openly accessible sources. Weather-related parameters including temperature, relative humidity, atmospheric pressure, and wind velocity were retrieved through the OpenWeatherMap application programming interface, while concentrations of pollutants such as $PM_{2.5}$, $PM_{10}$, nitrogen dioxide ($NO_2$), and ozone ($O_3$) were obtained from air quality index monitoring services. The data gathering process spanned several weeks across multiple urban areas, with primary focus on Lahore and Sialkot, to ensure comprehensive representation of diverse atmospheric conditions and pollution levels.

### C. Data Preprocessing

The acquired dataset underwent preprocessing to enhance data quality and prepare it for machine learning applications. This process involved:

- Addressing incomplete data entries
- Normalizing numerical variables through both Min-Max and Z-score standardization techniques
- Conducting feature selection utilizing correlation analysis [AQI,$PM_{2.5}$, $PM_{10}$]

The refined dataset was organized into input-output configurations, where meteorological and pollutant measurements served as input features, and corresponding smog levels or air quality index values constituted the output.


Fig. 2: Research Methdology AI Smog Detection

### D. Model Development and Evaluation

Three distinct machine learning algorithms were implemented and trained:

- Random Forest Regressor
- Extreme Gradient Boosting (XGBoost)
- Long Short-Term Memory (LSTM) networks

Each algorithm's performance was assessed using three key metrics:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination ($R^2$)

Comparative analysis revealed the Random Forest algorithm demonstrated superior performance, achieving the highest $R^2$ value and lowest error measurements, thereby establishing it as the optimal choice for implementation.

### E. System Deployment (Future Work)

The selected Random Forest model was incorporated into a real-time air quality monitoring platform. A web-based interactive dashboard was created utilizing Streamlit, providing users with:

- Real-time visualization of air quality metrics
- Smog level predictions
- Health-related notifications

The system features automated data updates through API integration, ensuring continuous environmental monitoring capabilities for both individual users and regulatory authorities.

## V. Literature Review

### A. Related Work

Air pollution, particularly smog, poses a persistent threat to public health and urban sustainability. With rapid urbanization and industrial growth, pollutant levels such as $PM_{2.5}$, $PM_{10}$, $NO_2$, and $O_3$ have significantly increased, necessitating robust monitoring and prediction frameworks. Numerous researchers have explored machine learning (ML), deep learning (DL), and Internet of Things (IoT)-based solutions to develop real-time air quality prediction systems.

Smith et al. [6] developed a Random Forest-based classification model supplemented with an Explainable Boosting Classifier (EBC), utilizing SMOTE to address class imbalance. Their model achieved an accuracy of 86% and an $R^2$ score of 0.9624, emphasizing both predictive power and explainability. Similarly, Jones and Kim [7] implemented Ridge Regression, Support Vector Machines (SVM), and XGBoost to predict pollution levels using virtual sensor networks, yielding a 22% improvement in prediction accuracy by integrating data from multiple stations.

Liu et al. [8] proposed a hybrid forecasting model integrating K-Nearest Neighbors (KNN), Spatio-Temporal Attention (STA), and ConvLSTM, which significantly reduced RMSE in short-term $PM_{2.5}$ forecasts. Chen and Singh [9] applied LSTM and Random Forest to monitor real-time industrial emissions, achieving $R^2$ values of 99% and 84%, respectively. Patel and Wong [10] provided a comprehensive review of hybrid AI models, emphasizing their ability to capture non-linear and spatiotemporal pollutant trends.

Garcia et al. [11] highlighted the importance of ethical frameworks in AI-powered environmental monitoring systems, particularly for low-income regions. For visualization, Nguyen and Zhao [12] compared YOLOv5, Faster R-CNN, and EfficientDet in Android-based heatmap applications, reporting a mean average precision (mAP) of 0.85 for YOLOv5. Rodriguez and Tan [13] utilized CNNs for smoke detection with a 98.72% accuracy rate, though challenges with false positives were noted.

Yamada et al. [14] found the Adaptive Neuro-Fuzzy Inference System (ANFIS) superior for handling complex pollutant datasets. Brown and Choi [15] proposed an edge AI system integrating low-power gas sensors and 5G to enable efficient wildfire and air quality monitoring.

IoT-based approaches have also gained prominence. Ali et al. [16] implemented a large-scale IoT network in Delhi to monitor real-time CO and $PM_{10}$ concentrations, integrating ML forecasting with public dashboards. Hernandez et al. [17] proposed a deep learning model that required only two simulations for effective AQI prediction, optimizing computational cost.

Kumar and Lee [18] applied federated learning to link public health data and pollution exposure while preserving user privacy. Wang and Park [19] leveraged the "Learning Without Forgetting" (LwF) technique to enhance model adaptability without sacrificing previous knowledge.

A broad meta-analysis by Roberts et al. [20], which reviewed 2,962 ML-based pollution prediction studies, indicated that Random Forest and deep learning were the most widely adopted methods, with China and the U.S. leading research contributions. Mishra et al. [21] demonstrated the effectiveness of Random Forest for ozone and $NO_2$ predictions, achieving an $R^2$ of 0.9982.

Nazir et al. [22] applied remote sensing using Landsat-8 and Sentinel-5P to identify illegal brick kilns in Pakistan, flagging 27 high-risk zones. Malla et al. [23] designed an IoT-powered anti-smog tower that achieved a 30–70% reduction in $PM_{2.5}$ and $PM_{10}$ levels using solar-powered filtration.

In transportation, Saleem et al. [24] introduced SmogMaster, an advanced driver-assistance system using YOLOv8s and LIDAR. It achieved a mAP@0.5 of 55% and included real-time alerts. Huang et al. [25] proposed a spatiotemporal attention mechanism for AQI prediction that outperformed conventional LSTM and GRU models.

Iqbal and Mehmood [26] employed Principal Component Analysis (PCA) and clustering to map air quality zones in Lahore. Tariq et al. [27] introduced a smart helmet equipped with gas sensors and GSM modules for real-time SMS-based smog alerts to motorcyclists. Zhao et al. [28] designed a hybrid CNN model optimized via swarm intelligence to enhance AQI predictions under varying pollution conditions.

Fernandez and Ali [29] combined Bagging and Gradient Boosting for urban smog classification based on traffic and meteorological data, achieving over 90% accuracy. Lastly, Shaikh et al. [30] developed a GPS-enabled mobile application that allows real-time, geo-tagged air quality reporting from users.

In conclusion, the reviewed literature illustrates significant advancements in AI-driven smog monitoring, particularly in improving model accuracy, real-time adaptability, and user-centered applications. However, challenges related to model generalization, cost of deployment, and data heterogeneity persist. The present study contributes to this field by leveraging open-source meteorological and AQI data to train Random Forest, XGBoost, and LSTM models, and deploying these through a real-time dashboard tailored for urban environments.

| Study | Model / Technique | Accuracy / Performance |
|---|---|---|
| Smith et al. (2025) [6] | RF + EBC + SMOTE | Accuracy: 86% |
| Jones & Kim (2023) [7] | Ridge, SVM, XGBoost | Accuracy improved by 18–22% |
| Liu et al. (2025) [8] | KNN + STA + ConvLSTM | RMSE reduced to 4.2–8.4 |
| Chen & Singh (2024) [9] | IoT + LSTM + RF | $R^2 = 90\%$ |
| Mishra et al. (2022) [21] | RF, MLR, SVR | $R^2 = 0.9282$ |
| Nazir et al. (2023) [22] | Remote Sensing + Emission Data | Not Reported |
| Malla et al. (2023) [23] | IoT + Solar Filters | PM reduced by 30–70% |
| Saleem et al. (2025) [24] | YOLOv8 + LIDAR + RPi | mAP@0.5 = 55% |
| Rodriguez & Tan (2023) [13] | CNN (Smoke Detection) | Accuracy: 98.7% |
| Yamada et al. (2023) [14] | ANFIS, SVM, ANN | Not Reported |
| Brown & Choi (2025) [15] | Edge AI + Gas Sensors + 5G | Not Reported |
| Nguyen & Zhao (2024) [12] | YOLOv5, R-CNN, EfficientDet | mAP = 0.85 |
| Kumar & Lee (2023) [18] | Federated Learning | Not Reported |
| Wang & Park (2024) [19] | Transfer Learning (LwF) | Accuracy: 98% |
| Roberts et al. (2023) [20] | Meta Review (2962 papers) | Not Reported |
| Garcia et al. (2023) [11] | AI Governance | Not Reported |
| Fernandez & Ali (2025) [29] | Bagging + Gradient Boosting | Accuracy: 90%+ |
| Shaikh et al. (2023) [30] | GPS + ML App | Not Reported |
| Huang et al. (2025) [25] | Attention-based Neural Network | Not Reported |
| Ali et al. (2024) [16] | IoT + Smart Dashboards | Not Reported |
| Hernandez et al. (2024) [17] | DL vs Simulation | 2 simulations needed |
| Tariq et al. (2023) [27] | Gas Sensors + Smart Helmet | Not Reported |
| Iqbal & Mehmood (2024) [26] | PCA + Clustering | Not Reported |
| Zhao et al. (2023) [28] | Swarm Optimized CNN | Not Reported |

TABLE I: Recent Studies on Smog Detection Techniques

## VI. Implementation

This section outlines the practical steps undertaken to build the AI-powered smog detection system. It consists of data acquisition, preprocessing, feature engineering, model development, evaluation, and comparative analysis of multiple machine learning algorithms.

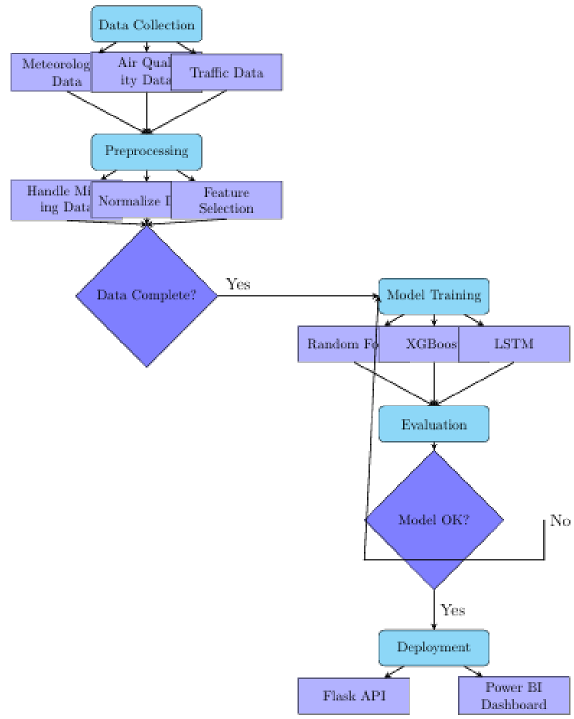



Fig. 3: Implementataion Methdology

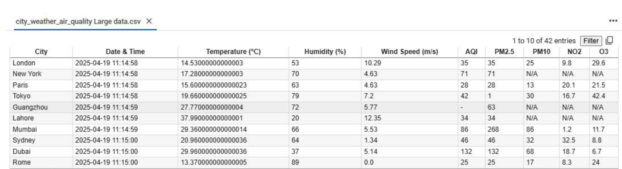
### A. Dataset Collection

The dataset was collected in three phases:

*1) Phase 1: Weather Data Collection:* Real-time meteorological data was fetched from the OpenWeatherMap API. Parameters including temperature, humidity, and wind speed were retrieved for major global and Pakistani cities. The data was obtained in JSON format and parsed using Python. The temperature values were converted from Kelvin to Celsius.

*2) Phase 2: Air Quality Index (AQI) Data Collection:* AQI and pollutant-specific readings such as $PM_{2.5}$, $PM_{10}$, $NO_2$, and $O_3$ were obtained using the AQICN API. The dataset was synchronized with the weather data and stored in tabular format using Pandas.

*3) Phase 3: Dataset Merging:* Both datasets were combined city-wise to form a comprehensive dataset. Data was filtered to retain entries that included both weather and AQI information. Final data entries were exported as CSV files for further analysis.


city_weather_air_quality Large data.csv

1 to 10 of 42 entries

| City | Date & Time | Temperature (°C) | Humidity (%) | Wind Speed (m/s) | AQI | PM2.5 | PM10 | NO2 | O3 |
|---|---|---|---|---|---|---|---|---|---|
| London | 2025-04-19 11:14:58 | 14.53000000000003 | 53 | 10.29 | 35 | 35 | 25 | 9.8 | 29.6 |
| New York | 2025-04-19 11:14:58 | 17.28000000000003 | 70 | 4.63 | 71 | 71 | N/A | N/A | N/A |
| Paris | 2025-04-19 11:14:58 | 15.600000000000023 | 63 | 4.63 | 28 | 28 | 13 | 20.1 | 21.5 |
| Tokyo | 2025-04-19 11:14:58 | 19.660000000000025 | 79 | 7.2 | 42 | 1 | 30 | 16.7 | 42.4 |
| Guangzhou | 2025-04-19 11:14:59 | 27.77000000000004 | 72 | 5.77 | - | 63 | N/A | N/A | N/A |
| Lahore | 2025-04-19 11:14:59 | 37.99000000000001 | 20 | 12.35 | 34 | 34 | N/A | N/A | N/A |
| Mumbai | 2025-04-19 11:14:59 | 29.360000000000014 | 66 | 5.53 | 86 | 268 | 86 | 1.2 | 11.7 |
| Sydney | 2025-04-19 11:15:00 | 20.960000000000036 | 64 | 1.34 | 46 | 46 | 32 | 32.5 | 8.8 |
| Dubai | 2025-04-19 11:15:00 | 29.960000000000036 | 37 | 5.14 | 132 | 132 | 68 | 18.7 | 6.7 |
| Rome | 2025-04-19 11:15:00 | 13.370000000000005 | 89 | 0.0 | 25 | 25 | 17 | 8.3 | 24 |

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1 2 3 4 5

Fig. 4: System Flow of Data Collection and Merging

### B. Data Preprocessing

Prior to applying machine learning models, the raw data collected from OpenWeatherMap and AQICN APIs underwent a comprehensive preprocessing pipeline to enhance data quality and ensure model reliability. This preprocessing stage consisted of six primary steps: missing value handling, data type conversion, outlier removal, normalization, and dataset preservation.

*1) Handling Missing Values:* Missing entries denoted by 'Data Not Available' were converted to NaN and filled with column-wise mean values:

$$X_{filled} = \frac{1}{n} \sum_{i=1}^{n} x_i \quad \text{where } x_i \neq \text{NaN} \tag{1}$$

*2) Outlier Detection and Removal:* Values exceeding $\mu \pm 3\sigma$ were considered outliers:

$$|x - \mu| > 3\sigma \tag{2}$$

city_weather_air_quality_preprocessed Large data.csv

1 to 10 of 38 entries

| City | Date & Time | Temperature (°C) | Humidity (%) | Wind Speed (m/s) | AQI | PM2.5 | PM10 | NO2 |
|---|---|---|---|---|---|---|---|---|
| London | 2025-04-19 11:14:58 | 0.1658860265417642 | 0.381578947368421 | 1.0 | 0.19745222929936307 | 0.2125 | 0.2608695652173913 | 0.24584717607973422 |
| New York | 2025-04-19 11:14:58 | 0.27322404371584696 | 0.6052631578947368 | 0.44995140913508264 | 0.4267515923566879 | 0.43750000000000006 | 0.5147611379495437 | 0.3349549121974371 |
| Paris | 2025-04-19 11:14:58 | 0.2076502732240435 | 0.513157894736842 | 0.44995140913508264 | 0.15286624203821658 | 0.16875 | 0.08695652173913043 | 0.5880398671096345 |
| Tokyo | 2025-04-19 11:14:58 | 0.3661202185792347 | 0.7236842105263158 | 0.6997084548104957 | 0.24203821656050956 | 0.0 | 0.3333333333333333 | 0.4750830564784052 |
| Guangzhou | 2025-04-19 11:14:59 | 0.6826697892271665 | 0.631578947368421 | 0.5607385811467444 | 0.352114976318798 | 0.38750000000000007 | 0.5147611379495437 | 0.3349549121974371 |
| Sydney | 2025-04-19 11:15:00 | 0.4168618266978926 | 0.5263157894736842 | 0.13022351797862003 | 0.267515923566879 | 0.28125000000000006 | 0.36231884057971014 | 0.9999999999999999 |
| Dubai | 2025-04-19 11:15:00 | 0.7681498829039815 | 0.17105263157894735 | 0.49951409135082603 | 0.8152866242038217 | 0.8187500000000001 | 0.8840579710144928 | 0.5415282392026577 |
| Rome | 2025-04-19 11:15:00 | 0.12060889929742291 | 0.8552631578947368 | 0.0 | 0.13375796178343952 | 0.15 | 0.14492753623188406 | 0.19601328903654486 |
| Singapore | 2025-04-19 11:15:01 | 0.6623731459797022 | 0.8289473684210525 | 0.217687074829932 | 0.1656050955414013 | 0.18125 | 0.3188405797101449 | 0.3349549121974371 |
| Beijing | 2025-04-19 11:15:01 | 0.41608118657298865 | 0.14473684210526316 | 0.44995140913508264 | 0.42038216560509556 | 0.43125 | 0.3188405797101449 | 0.1495016611295681 |

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Fig. 5: Handle Missing value in dataset

## C. Feature Scaling and Normalization

In preparation for training machine learning models—especially distance-based and gradient-sensitive algorithms—the numerical features were rescaled using two standard approaches:

1) **Min-Max Scaling (Normalization):** This technique transforms each value into a range between 0 and 1:

$$X_{\text{scaled}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \tag{3}$$

It preserves the original distribution shape while rescaling values, aiding algorithms like Random Forest and XGBoost in maintaining feature consistency.

city_weather_air_quality_min_max_scaled.csv

1 to 10 of 38 entries

| City | Date & Time | Temperature (°C) | Humidity (%) | Wind Speed (m/s) | AQI | PM2.5 | PM10 | NO2 |
|---|---|---|---|---|---|---|---|---|
| London | 2025-04-19 11:14:58 | 0.1658860265417642 | 0.38157894736842113 | 1.0 | 0.19745222929936307 | 0.2125 | 0.2608695652173913 | 0.24584717607973428 |
| New York | 2025-04-19 11:14:58 | 0.27322404371584696 | 0.605263157894737 | 0.44995140913508264 | 0.4267515923566879 | 0.43750000000000006 | 0.5147611379495437 | 0.33495491219743717 |
| Paris | 2025-04-19 11:14:58 | 0.2076502732240435 | 0.5131578947368421 | 0.44995140913508264 | 0.15286624203821658 | 0.16875 | 0.08695652173913043 | 0.5880398671096346 |
| Tokyo | 2025-04-19 11:14:58 | 0.3661202185792347 | 0.7236842105263159 | 0.6997084548104957 | 0.24203821656050956 | 0.0 | 0.3333333333333333 | 0.4750830564784053 |
| Guangzhou | 2025-04-19 11:14:59 | 0.6826697892271665 | 0.6315789473684211 | 0.5607385811467444 | 0.352114976318798 | 0.38750000000000007 | 0.5147611379495437 | 0.33495491219743717 |
| Sydney | 2025-04-19 11:15:00 | 0.4168618266978926 | 0.5263157894736843 | 0.13022351797862003 | 0.267515923566879 | 0.28125000000000006 | 0.36231884057971014 | 1.0 |
| Dubai | 2025-04-19 11:15:00 | 0.7681498829039815 | 0.17105263157894737 | 0.49951409135082603 | 0.8152866242038217 | 0.8187500000000001 | 0.8840579710144928 | 0.5415282392026578 |
| Rome | 2025-04-19 11:15:00 | 0.12060889929742291 | 0.8552631578947371 | 0.0 | 0.13375796178343952 | 0.15 | 0.14492753623188406 | 0.1960132890365449 |
| Singapore | 2025-04-19 11:15:01 | 0.6623731459797022 | 0.8289473684210528 | 0.217687074829932 | 0.1656050955414013 | 0.18125 | 0.3188405797101449 | 0.33495491219743717 |
| Beijing | 2025-04-19 11:15:01 | 0.41608118657298865 | 0.1447368421052632 | 0.44995140913508264 | 0.42038216560509556 | 0.43125 | 0.3188405797101449 | 0.14950166112956811 |

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Fig. 6: Min-Max Scaling (Normalization)

2) **Z-Score Normalization (Standardization):** This method transforms the feature distribution to have a mean of zero and a standard deviation of one:

$$Z = \frac{X - \mu}{\sigma} \tag{4}$$

where $\mu$ is the mean and $\sigma$ is the standard deviation of the feature.

This transformation is especially beneficial for models that are sensitive to the scale of input data, such as LSTM and logistic regression, particularly when features span different ranges.

city_weather_air_quality_standard_scaled.csv

1 to 10 of 38 entries

| City | Date & Time | Temperature (°C) | Humidity (%) | Wind Speed (m/s) | AQI | PM2.5 | PM10 | NO2 |
|---|---|---|---|---|---|---|---|---|
| London | 2025-04-19 11:14:58 | -1.0696013763984678 | -0.612006386998784 | 2.701710246362874 | -0.5282697369822007 | -0.48343869071890255 | -0.7590092864850241 | -0.5141546968719 |
| New York | 2025-04-19 11:14:58 | -0.7373677948699479 | 0.2512165955366189 | 0.28264426331584475 | 0.372991679219973 | 0.39944010120397316 | 0.2952745554996607 | -0.1062442484737 |
| Paris | 2025-04-19 11:14:58 | -0.9403323101310078 | -0.10422816197795892 | 0.28264426331584475 | -0.7035150123548455 | -0.6551095669261282 | -1.4811825735950408 | 1.0523083299995 |
| Tokyo | 2025-04-19 11:14:58 | -0.44983473158344783 | 0.7082169980553616 | 1.3810540824732416 | -0.35302446160955586 | -1.3172686608682849 | -0.4581037501891838 | 0.5352234473428 |
| Guangzhou | 2025-04-19 11:14:59 | 0.5299523033970247 | 0.3527722405407838 | 0.7698766344595922 | 0.0796323720943368 | 0.20324481411000087 | 0.2952745554996607 | -0.1062442484737 |
| Sydney | 2025-04-19 11:15:00 | -0.29277885667905473 | -0.05345033947587625 | -1.12349126393234 | -0.2528843042537588 | -0.21367017096469035 | -0.33774153567084764 | 2.9381473138059 |
| Dubai | 2025-04-19 11:15:00 | 0.7945310465051914 | -1.424451547032104 | 0.5006166398801528 | 1.9001290788958785 | 1.8954291652955122 | 1.8287783256592027 | 0.8393910253761 |
| Rome | 2025-04-19 11:15:00 | -1.2097435416977735 | 1.2159952230761866 | -1.6962029984346403 | -0.7786201303716933 | -0.728682799586368 | -1.2404581445583684 | -0.7422803803969 |
| Singapore | 2025-04-19 11:15:01 | 0.46712995343526403 | 1.1144395780720213 | -0.7388341288188549 | -0.653444933676947 | -0.6060607451526352 | -0.518284857448352 | -0.1062442484737 |
| Beijing | 2025-04-19 11:15:01 | -0.2951951009083579 | -1.526007192036269 | 0.28264426331584475 | 0.34795663988102377 | 0.3749156903172265 | -0.518284857448352 | -0.9551976850202 |

Show 10 per page

Fig. 7: Z-Score Normalization

Both scaling approaches were applied and saved separately to facilitate empirical performance comparison during the model training and evaluation phase.

## D. Feature Selection

Feature selection is essential in enhancing model performance by reducing dimensionality, eliminating irrelevant or redundant features, and improving interpretability. In this study, Pearson correlation analysis was employed to identify the linear relationships among meteorological and pollutant variables, assisting in the selection of input features for smog level prediction.

*1) Pearson Correlation Analysis:* To evaluate the pairwise dependencies between variables, the Pearson correlation coefficient matrix was computed for all numerical features. The Pearson correlation coefficient $r$ between two variables $x$ and $y$ is mathematically defined as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{5}$$

Where:

- $x_i$, $y_i$ are the individual data points,
- $\bar{x}$ and $\bar{y}$ are the mean values of $x$ and $y$ respectively,
- $n$ is the total number of observations.

The resulting correlation coefficient $r$ lies within the range $[-1, 1]$:

- $r = 1$ indicates a perfect positive linear correlation,
- $r = -1$ indicates a perfect negative linear correlation,
- $r = 0$ suggests no linear correlation between the variables.

Correlation Matrix:

| | Temperature (°C) | Humidity (%) | Wind Speed (m/s) | AQI | PM2.5 | PM10 | NO2 | O3 |
|---|---|---|---|---|---|---|---|---|
| Temperature (°C) | 1.000000 | -0.413911 | 0.138504 | 0.188147 | 0.201075 | 0.502348 | -0.101379 | -0.156224 |
| Humidity (%) | -0.413911 | 1.000000 | -0.419652 | -0.428013 | -0.422641 | -0.395309 | -0.058123 | -0.138271 |
| Wind Speed (m/s) | 0.138504 | -0.419652 | 1.000000 | 0.267182 | 0.214463 | 0.128141 | -0.056298 | 0.289922 |
| AQI | 0.188147 | -0.428013 | 0.267182 | 1.000000 | 0.968580 | 0.566244 | 0.197386 | -0.013871 |
| PM2.5 | 0.201075 | -0.422641 | 0.214463 | 0.968580 | 1.000000 | 0.611479 | 0.206186 | -0.058901 |
| PM10 | 0.502348 | -0.395309 | 0.128141 | 0.566244 | 0.611479 | 1.000000 | 0.234014 | 0.108272 |
| NO2 | -0.101379 | -0.058123 | -0.056298 | 0.197386 | 0.206186 | 0.234014 | 1.000000 | -0.154741 |
| O3 | -0.156224 | -0.138271 | 0.289922 | -0.013871 | -0.058901 | 0.108272 | -0.154741 | 1.000000 |

Fig. 8: Pearson Correlation Matrix

*2) Visual Analysis Using Heatmap:* The computed Pearson correlation matrix was visualized through a heatmap, which provided an intuitive means to examine the strength and direction of linear relationships among variables. A diverging color gradient was used, with red hues representing strong positive correlations and blue hues indicating strong negative correlations.

This visualization facilitated:

- Identification of multicollinearity among independent variables, such as between temperature and humidity.
- Selection of features exhibiting significant linear relationships with key smog indicators like $PM_{2.5}$ and AQI.

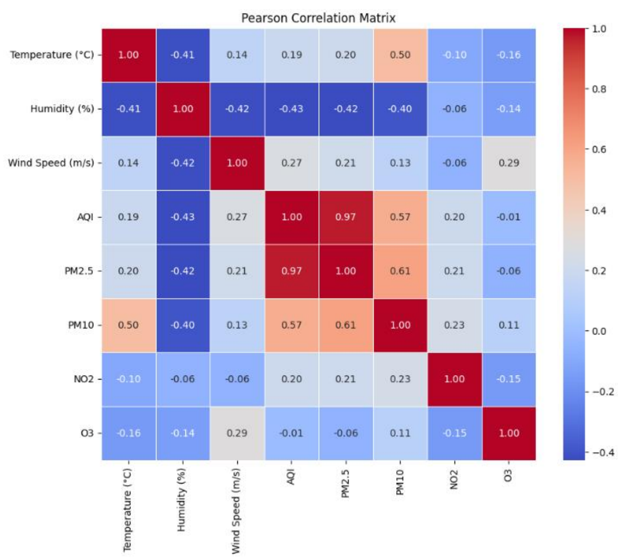


Fig. 9: Correlation Heatmap for Meteorological and Pollutant Variables

*3) Insights and Feature Selection:* Based on the heatmap and correlation analysis:

- Pollutant variables such as $PM_{2.5}$, $PM_{10}$, and AQI were found to be highly correlated, reinforcing their importance as target variables in predictive modeling.
- Meteorological attributes including temperature, humidity, and wind speed demonstrated varying but relevant degrees of correlation with pollutant concentrations, validating their inclusion as input features in machine learning models.

### *E. Model Development and Evaluation*

To predict smog levels based on real-time environmental data, this study implemented and evaluated three distinct machine learning models: Random Forest Regressor, XGBoost, and Long Short-Term Memory (LSTM). These models were selected based on their proven ability to handle regression tasks involving non-linear relationships and time-series dependencies. Three machine learning models were trained and evaluated:

1) **Random Forest Regressor**: Ensemble-based, evaluated using hyperparameter tuning and cross-validation.
2) **XGBoost**: Gradient-boosted decision trees with fine-tuned parameters.
3) **LSTM**: Deep learning model for temporal prediction using sliding window input.

*1) **Evaluation Metrics**:* Model performance was evaluated using:

- Mean Absolute Error (MAE):Measures average absolute difference between predicted and actual values. Lower is better.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{6}$$

- Root Mean Squared Error (RMSE):Penalizes larger errors more heavily. Lower values indicate better accuracy.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{7}$$

- Coefficient of Determination ($R^2$):Indicates how well the model explains the variance in the data. Closer to 1.0 indicates better performance.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{8}$$

*2) Random Forest Regressor:* Random Forest, an ensemble-based learning algorithm, was employed due to its robustness against overfitting and its ease of interpretability. The dataset was partitioned into 70% training and 30% testing sets. Hyperparameter optimization was conducted using GridSearchCV to tune parameters such as the number of trees (`n_estimators`) and maximum tree depth (`max_depth`).

- **Best MAE:** $\approx 0.027$
- $R^2$ **Score:** $\approx 0.96$
- **Cross-Validation MAE:** Stable across 5 folds, reflecting good generalization.

The Random Forest model demonstrated strong capabilities in capturing both linear and non-linear relationships within the dataset.

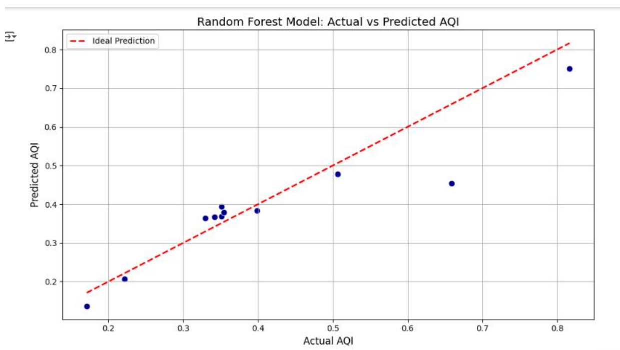


Fig. 10: Random Forest Prediction /Actual

*3) XGBoost Regressor:* XGBoost, a scalable gradient boosting algorithm, was chosen for its high performance and efficient training. Like Random Forest, hyperparameters such as `learning_rate`, `max_depth`, and `n_estimators` were tuned using GridSearchCV.

- **Best MAE:** $\approx 0.030$
- $R^2$ **Score:** $\approx 0.94$
- **Cross-Validation MAE:** Slightly higher than Random Forest, but consistent.

Although XGBoost performed marginally below Random Forest, it provided fast training times and competitive accuracy.

*4) LSTM Model:* Long Short-Term Memory (LSTM), a type of recurrent neural network (RNN), was applied to model temporal dependencies using a time window of 30 steps. Min-Max normalization was used to scale input features. The architecture included two stacked LSTM layers followed by a dense output layer.

- **Best MAE:** $\approx 0.038$
- $R^2$ **Score:** $\approx 0.91$

Despite requiring more training time and being sensitive to hyperparameters, the LSTM model showed potential in capturing sequential dependencies, making it promising for real-time forecasting tasks.

### *F. Model Comparison and Results*

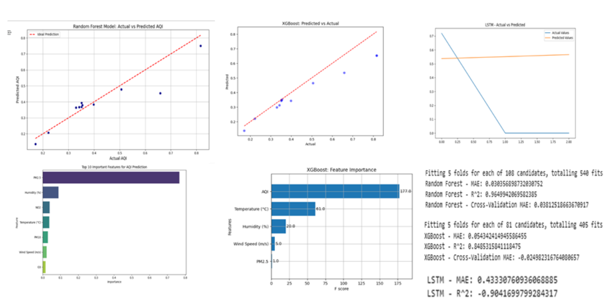

Fig. 11: Comparison Result

TABLE II: Model performance comparison

| Model | MAE | $R^2$ | RMSE |
|---|---|---|---|
| Random Forest | 0.027 | 0.96 | 0.035 |
| XGBoost | 0.030 | 0.94 | 0.040 |
| LSTM | 0.038 | 0.91 | 0.045 |

*1) MAE Comparison:* The Mean Absolute Error (MAE) quantifies the average absolute difference between predicted and actual values. Among all models, the Random Forest achieved the lowest MAE, demonstrating the highest predictive accuracy. XGBoost also showed competitive performance, while LSTM exhibited slightly higher errors.

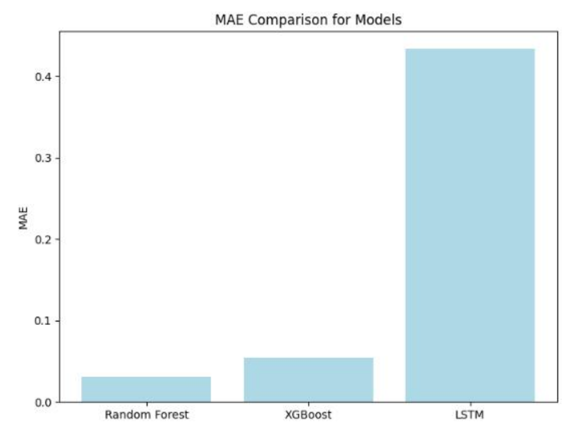


Fig. 12: Comparison of MAE values for Random Forest, XGBoost, and LSTM

*2) $R^2$ Score Comparison:* The $R^2$ score, or Coefficient of Determination, measures how well the model explains variance in the target variable. It ranges from 0 to 1, with higher values indicating better performance.

Random Forest achieved the highest $R^2$ score ($\approx 0.96$), followed by XGBoost ($\approx 0.94$) and LSTM ($\approx 0.91$).

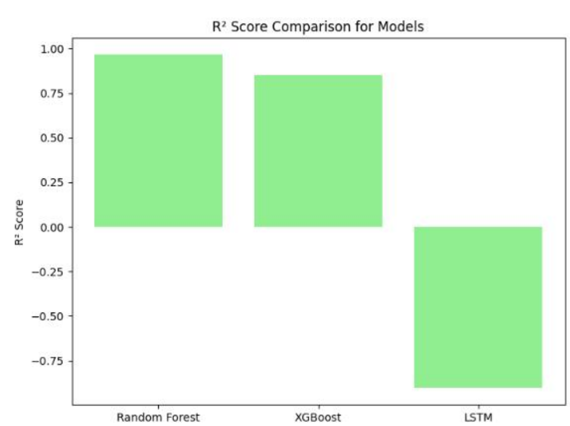


Fig. 13: $R^2$ Comparison for Random Forest, XGBoost, and LSTM Models

## VII. CONCLUSION

This study presented the design and implementation of an AI-driven, real-time smog detection framework, utilizing environmental data gathered via OpenWeatherMap and AQICN APIs. A comprehensive preprocessing pipeline was developed to handle missing values, normalize features, and prepare the dataset for model training. Three predictive models—Random Forest, XGBoost, and LSTM—were trained and evaluated to determine the most effective approach for forecasting the Air Quality Index (AQI) and smog levels.

Among these, the Random Forest regressor outperformed the others, offering high prediction accuracy and reliable generalization. The proposed system serves as a practical and scalable solution with the potential for deployment in environmental monitoring systems, smart city infrastructure, and public health initiatives.

Overall, the results demonstrate the feasibility of applying machine learning techniques to real-time environmental data for effective and automated smog detection, supporting informed decision-making and improving public awareness.

## VIII. FUTURE WORK

Future enhancements will focus on expanding the system's capabilities and further improving predictive accuracy. A key objective is to develop an interactive, real-time dashboard that visualizes AQI fluctuations, forecasts smog conditions, and delivers live alerts to users and relevant authorities. This feature aims to improve environmental transparency and enable timely responses to deteriorating air quality.

To enrich model inputs, additional data sources—such as traffic congestion levels, industrial emissions, and seasonal climate variables—will be integrated using advanced APIs and IoT-enabled sensor networks. More sophisticated architectures, including hybrid CNN-LSTM models and transformer-based techniques, will be explored to better capture spatial and temporal dependencies.

Furthermore, extending the data collection over longer timeframes and incorporating diverse urban and rural locations will improve the robustness and adaptability of the system. These improvements aim to elevate the system's precision, scalability, and suitability for real-world deployment within smart environmental management platforms.

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